

CSMFAB000000000022 Smart Rope Actuator Design Specification

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | Date: June 2026 Category: Space Launch / Actuator

Revision Drivers (all V2.0 documents)

- 1. 2025-2026 experimental and regulatory data supersedes V1.0 specifications
- 2. Solar Cycle 25 SSN ~137 peak (observed) — elevated urgency for all Carrington-focused work
- 3. Standards updates — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
Piezoelectric actuator material	PZT-5H (lead, Tc=193degC)	KNbO3-BaTiO3 (lead-free, Tc=310degC)	RoHS violation; thermally unreliable at -60degC altitude
Hinge material	Spring steel (ferromagnetic)	Ti3AlC2 MAX Phase (non-magnetic, machinable)	GIC failure at altitude; brittle at -60degC
EMI shield	Cu mesh SE=45 dB	MXene Ti3C2Tx SE=92 dB	47 dB shielding deficit in DEW/HPM environment
Actuator temp range	-50degC to +200degC	-60degC to +400degC	V1.0 actuator fails at 30km altitude temperature
V2.0 velocity estimate	~250 m/s estimated	320-450 m/s (DE actuator data at altitude)	V1.0 velocity was under-estimated

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMFAB000000000022 Smart Rope Actuator Design Specification | June 2026